

Total No. of Questions : 8]

SEAT No. :

PD1711

[Total No. of Pages : 3

[6432]-404

S.Y.M.C.A. (Management)

MT-41 : OPTIMIZATION TECHNIQUES

(2019 Pattern) (Semester -IV)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Question No 1 is compulsory.*
- 2) *Attempt any five questions from Question no 2 to question No. 8.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of Non-programmable calculator & statistical table is allowed.*

Q1) a) The following table shows the jobs of a network along with their time estimates **[10]**

Jobs	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)
1-2	1	7	13
1-6	2	5	14
2-3	2	14	26
2-4	2	5	8
3-5	7	10	19
4-5	5	5	17
6-7	5	8	29
5-8	3	3	9
7-8	8	17	32

- i) Draw the network diagram
- ii) Find the critical path
- iii) Project completion time and project variance
- iv) Probability that project will be completed in 40 days

P.T.O.

- b) Solve the following transportation problem to find optimum solution and cost associated with it [10]

Sources	Destination				Supply
	1	2	3	4	
1	11	13	17	14	250
2	16	18	14	10	300
3	21	24	13	10	400
Demand	200	225	275	250	

- Q2) Solve the following game. Find the optimal strategy of A and B, value of the game. The pay off matrix is as given below [10]

		Player B				
		B ₁	B ₂	B ₃	B ₄	B ₅
Player A	A ₁	9	3	1	8	0
	A ₂	6	5	4	6	7
	A ₃	2	4	3	3	8
	A ₄	5	6	2	2	1

- Q3) Annual demand of an item is 2,400 units ordering cost is Rs 350 per order. Inventory carrying cost is 24% of the purchase price per year. Price per item is 9.25 Rs. Find [10]

- Economic order quantity
- Total cost related to economic order quantity
- Optimum number of orders
- Time between two orders

- Q4) A company has 4 machines and 4 tasks The time required for each task on each machine is given below (Time in hours). Find the assignment which minimize the total set up time for all tasks [10]

	Task 1	Task 2	Task 3	Task 4
Machine 1	13	8	7	6
Machine 2	7	11	5	4
Machine 3	6	7	6	8
Machine 4	5	6	5	9

Q5) Customer arrive at a one window drive through pharmacy according to poisson distribution with a mean of 10 per hour. The service time per customer is exponential with a mean of 5 minutes **[10]**

- What is the probability that customer has to wait
- Number of customers in the system
- Average waiting time of customer before getting the service

Q6) Solve the following using simplex method. **[10]**

$$\begin{aligned} \text{Max } Z &= 3x_1 + 2x_2 + 5x_3 \\ \text{subject to } x_1 + x_2 + x_3 &\leq 9 \\ 2x_1 + 3x_2 + 5x_3 &\leq 20 \\ 2x_1 - x_2 - x_3 &\leq 8 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

Q7) A Machine owner finds from his past records that the maintainance costs per year of a machine whose purchase price is Rs. 8,000 are as given below.

Year	1	2	3	4	5	6	7	8
Maintainance cost	1,000	1,200	1,700	2,200	2,900	3,800	4,800	6,000
Resale value	4,000	2,000	1,200	600	500	400	400	400

Determine at which time it is profitable to replace the machine **[10]**

Q8) Six jobs go first on macine A, then machine B & then machine C. The time of their processing on 3 machine in hours is given below **[10]**

Jobs	1	2	3	4	5	6
Machine A	8	3	7	2	5	2
Machine B	3	4	5	2	1	6
Machine C	8	7	6	9	10	9

Find the sequence of jobs that minimizes total elapsed time to complete all jobs Also find idle time of all 3 machines

